LECTURE NOTES

Computer Networks

Network Types

PAN & HAN | LAN & CAN | MAN & WAN | EPN | VPN | SAN

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2 hrs Theory

Where We Left Off in Week 2

We built networks from devices. Now we ask: what kind of network did we build?

Week 2: hardware (NIC, switch, router…), software (drivers, stacks, services), IP addressing, connection types, and physical topologies (bus, star, ring, mesh, tree, hybrid).

We chose a topology

Topology is the shape. But a network also has a size and an owner — those are what we classify today.

→ The two axes, next

We mentioned PAN to WAN

Week 1 named the size tiers in one line. This week we give each one a proper treatment.

→ Parts 1 to 3

We addressed devices privately

Private addresses hinted at private networks. VPNs make that idea travel across the Internet.

→ Part 4: VPN

We connected servers

A server needs storage. When storage gets its own network, it becomes a SAN.

→ Part 5: SAN

Session Roadmap

Networks classified two ways: by how far they reach, and by who owns them

01

The Two Axes

Geographic scope and ownership

15 min

02

Small Scope

PAN, HAN, and LAN

25 min

03

Large Scope

CAN, MAN, and WAN

25 min

04

Ownership: EPN & VPN

Private networks, and private over public

25 min

05

Purpose: SAN

A network built only for storage

15 min

06

Practical

Classify and simulate multi-site networks

60 min

LESSON

Weekly Learning Outcomes

By the end of this session you will be able to:



01

Classify a Network Scenario

Given a real situation, name the network type by its geographic scope and its ownership.



02

Explain the Purpose of a VPN

Say what problem a VPN solves, how a tunnel works, and where organisations use one.



03

Distinguish a SAN from a LAN

Explain how a storage area network differs from an ordinary LAN in purpose and design.

Keep asking two questions: how far does this network reach, and who owns and controls it? Those two answers name almost every network type.

01

The Two Axes of Classification

Before the names, understand the two questions that generate all of them

Geographic scope

Ownership & control

Why Do We Classify Networks?

The type tells you what to expect before you see a single cable

“Network” covers everything from two earbuds talking to a phone, up to the entire global Internet. Those are not the same engineering problem. Classifying a network tells you its likely speed, its cost, who is responsible for it, and what can go wrong — all from a three-letter label.

How far it reaches

A few metres, a building, a city, or the planet.

How fast it is

Short distances allow much higher speeds than long ones.

Who owns it

You, one organisation, or a telecom carrier you rent from.

What it costs

A cable you buy once, or a monthly line you lease forever.

The core trade-off: as a network grows in reach, it almost always drops in speed and rises in cost. Distance is never free.

Axis 1: Geographic Scope

Most network types are named after the area they cover

PAN
~1–10 m

Personal Area Network

Phone + earbuds + smartwatch

LAN
One building / site

Local Area Network

A computer lab, a home, an office floor

CAN
A few buildings

Campus Area Network

A university or a hospital campus

MAN
A city

Metropolitan Area Network

A council linking offices across a city

WAN
Countries, the world

Wide Area Network

The Internet; a bank across continents

Notice: HAN (home) is really a LAN in a house, and EPN is a private WAN. The size tiers are the backbone; the others are special cases.

Axis 2: Ownership and Control

Who owns the wires, and who is allowed to use them?

Two networks can be the same size but completely different in who runs them. A LAN is usually owned outright by one organisation. A WAN is usually rented from a telecom carrier, because no single company can lay cable across a country. This ownership question generates a second family of names.

Private

Owned and used by one organisation. Access is restricted to its own people.

Your home LAN · a company EPN

Public

Shared infrastructure that anyone may use, usually run by carriers.

The Internet · public Wi-Fi

Private over Public

A private, encrypted path carried across public infrastructure.

A VPN tunnel over the Internet

Special-purpose

A network built and reserved for one job, not general traffic.

A SAN, for storage only

The whole week in one sentence: scope tells you how big, ownership tells you whose. Every network type is just a point on those two axes.

02

Small-Scope Networks

The networks around one person, one home, and one building

PAN

HAN

LAN

Personal Area Network (PAN)

The smallest network there is — the one that travels with you

PAN

A PAN connects the personal devices of one individual, within a few metres of their body. It usually forms and dissolves automatically as devices come into range, with no configuration by the user at all.

At a glance

Range	About 1 to 10 metres
Owner	One individual person
Technology	Bluetooth, NFC, USB, Zigbee
Speed	Low — kbps to a few Mbps
Setup	Automatic pairing, no admin

Key characteristics

- Wearables: watch, fitness band, earbuds
- Centres on a phone or laptop
- Very low power, by design
- Forms and breaks up on the move
- Short range is its main security

Everyday example: your phone connected to wireless earbuds and a smartwatch at the same time is a working PAN — three devices, no cables, no setup.

Home Area Network (HAN)

A LAN that happens to live inside a house

HAN

A HAN connects all the digital devices inside a single home so they can share one Internet connection and reach each other. It is technically a small LAN, but the name is used when the setting is domestic and includes smart-home devices.

At a glance

Range	One house or apartment
Owner	The household
Technology	Wi-Fi, Ethernet, Zigbee
Speed	Tens to hundreds of Mbps
Centre	The home router / gateway

Key characteristics

- PCs, phones, TVs, game consoles
- Smart-home: lights, thermostat, plugs
- Shares one broadband connection
- Mostly wireless in practice
- Managed by a non-expert user

Everyday example: your laptop, phone, smart TV, and a smart bulb all joined to the home router, sharing one Internet line — that is a HAN.

Local Area Network (LAN)

The workhorse: one building, one owner, high speed

LAN

A LAN connects devices within a single, limited area such as an office, a lab, or a floor of a building. It is owned entirely by one organisation, which means it can be fast, cheap to run, and fully under that organisation's control.

At a glance

Range	A room to a building
Owner	One organisation, outright
Technology	Ethernet, Wi-Fi
Speed	High — 1 to 10 Gbps typical
Cost	Low running cost once built

Key characteristics

- Built around switches, in a star
- Very high speed, very low delay
- You own it, so no monthly line rental
- Fully under the owner's control
- Easy to extend within the site

Everyday example: this computer lab is a LAN: 40 PCs, one switch, one owner — the university — all inside one room, running at gigabit speed.

03

Large-Scope Networks

When a network outgrows a single building

CAN

MAN

WAN

Campus Area Network (CAN)

Several nearby buildings, linked into one network by their owner

CAN

A CAN connects the LANs of several buildings that sit close together on land owned by one organisation — a university, a hospital, or a corporate campus. Because the owner controls the ground between the buildings, they can lay their own fiber and keep LAN-like speeds across the whole site.

At a glance

Range	A few adjacent buildings
Owner	One organisation
Technology	Fiber backbone + Ethernet
Speed	High — near-LAN across the site
Made of	Many LANs joined by routers

Key characteristics

- Links building to building
- Routers join each building's LAN
- Owner-laid fiber, so no line rental
- Central IT controls the whole campus
- Classic case: a university

Right here: Rajarata University's network is a CAN — the Computing, Science, and other faculty LANs, joined by a campus fiber backbone under one IT department.

Metropolitan Area Network (MAN)

A network the size of a city

MAN

A MAN spans a city or a large town, connecting multiple sites that are too far apart for a CAN but still within one metropolitan area. It is often built and run by a city government or a telecom provider, and frequently rides on a ring of optical fiber around the city.

At a glance

Range	A city — up to ~50 km
Owner	City body or a carrier
Technology	Fiber, Metro Ethernet
Speed	High, but below a LAN
Shape	Often a fiber ring

Key characteristics

- Connects sites across a city
- City council, hospitals, campuses
- Ring design survives one break
- Bridges many LANs to the Internet
- May be leased from a provider

Everyday example: a city council linking its town hall, libraries, and branch offices across Colombo over a fiber ring it owns is a MAN.

Wide Area Network (WAN)

The largest scope — across countries, continents, and the whole planet

WAN

A WAN connects sites separated by large distances — different cities, countries, or continents. No single organisation can lay cable that far, so a WAN almost always runs over infrastructure leased from telecom carriers. The Internet is the largest WAN of all.

At a glance

Range	Countries to global
Owner	Usually leased from carriers
Technology	Fiber, MPLS, leased lines, satellite
Speed	Lower than a LAN; high latency
Cost	High — recurring line rental

Key characteristics

- Links geographically distant sites
- You rent it; you rarely own it
- Long distance means real latency
- Routers pick paths hop by hop
- The Internet is the biggest WAN

Everyday example: a bank connecting its Colombo head office to branches in London and Singapore uses a WAN — leased links spanning continents.

How WANs Are Actually Built

You do not lay your own cable across an ocean — you rent one of these

Leased Line

A dedicated point-to-point circuit rented from a carrier, reserved for you alone.

→ *Private, reliable, expensive*

MPLS

A carrier service that label-switches your traffic across its network with agreed performance.

→ *Common for business WANs*

The Internet

The cheapest option: send traffic over the public Internet, usually secured with a VPN.

→ *Cheap, but best-effort*

Satellite

Links remote sites with no cable access, at the cost of high latency.

→ *Reaches anywhere; slow*

This is why VPNs exist: the Internet is by far the cheapest WAN, but it is public. A VPN makes it private enough to trust — our next major topic.

Scope Networks Side by Side

The same five tiers, compared on what actually changes with distance

Type	Range	Typical owner	Technology	Speed	Wires
PAN	~1–10 m	One person	Bluetooth, USB	Low	Owned
LAN	One building	One organisation	Ethernet, Wi-Fi	Very high	Owned
CAN	A few buildings	One organisation	Fiber + Ethernet	High	Owned
MAN	A city (~50 km)	City or carrier	Metro fiber	High	Mixed
WAN	Countries, global	Carriers	MPLS, leased, satellite	Lower, high latency	Leased

Read down the Speed column: it falls as range grows. That single trade-off — reach versus speed — explains the whole table.

04

Ownership: EPN and VPN

Networks defined not by size, but by who is allowed inside them

Enterprise Private Network

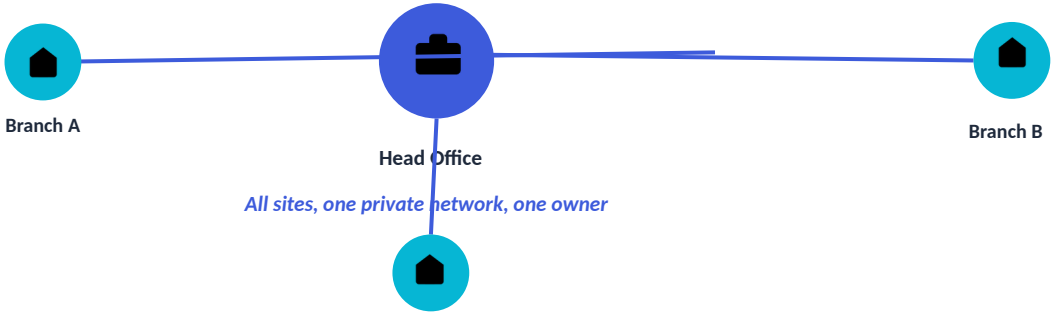
Virtual Private Network

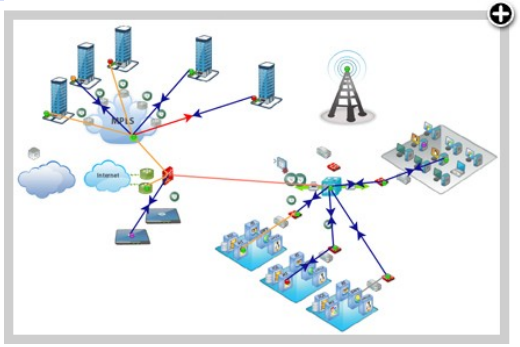
Enterprise Private Network (EPN)

One company's own private network, spanning all of its sites

EPN

An EPN is a network built and controlled by a single organisation to securely connect all of its locations. It is private in ownership: the company builds it for its own exclusive use, often across many cities, so it is effectively a private WAN.





Private and exclusive

Only the organisation's own traffic ever uses it.

High control and security

The company sets every policy end to end.

Expensive to build

Leased lines between sites cost a great deal every month.

The Problem a VPN Solves

You want the privacy of an EPN, but not its cost

An EPN with leased lines is secure but expensive. The Internet is cheap but public — anyone along the path could, in principle, read your traffic. A VPN gives you a third option: use the cheap public Internet, but wrap your traffic so that it is as private as a leased line.



Leased Line (EPN)

Privacy: **Very private**

Cost: **Very expensive**

Secure but costly



Plain Internet

Privacy: **Not private**

Cost: **Very cheap**

Cheap but exposed



VPN over Internet

Privacy: **Private (encrypted)**

Cost: **Cheap**

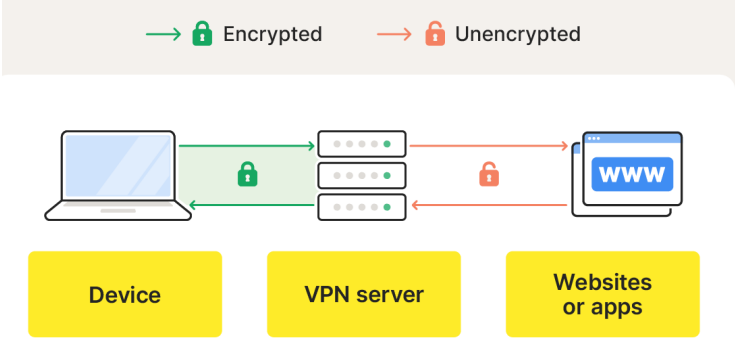
The best of both

Definition: a VPN is a secure, encrypted connection carried across a public network, making a shared path behave like a private one.

How a VPN Tunnel Works

Encapsulation and encryption, working together

A VPN wraps each of your private packets inside a new, encrypted outer packet. The public Internet only sees the outer wrapper and where it is going; it cannot read the private packet inside. At the far end, the wrapper is removed and the original packet continues. This wrapped path is called a tunnel.



What the wrapping looks like

New public header	Your original private packet — encrypted, unreadable from outside

Encapsulation
wraps the private packet inside a public one

Encryption
scrambles the contents so only the ends can read it

Authentication
proves each end is who it claims to be

Where VPNs Are Used

Two common shapes, and the reasons organisations and individuals rely on them

Remote-Access VPN

One person connects securely into their organisation from outside — from home, a hotel, or a café.

Example: a lecturer marking from home over the university VPN.

Site-to-Site VPN

Two whole offices are joined over the Internet, as if they shared one private network.

Example: a branch office linked to head office without a leased line.

Why organisations and people use them

Save money

Replace costly leased lines with the cheap public Internet.

Enable remote work

Staff reach internal systems safely from anywhere.

Protect data

Encryption shields traffic on untrusted public Wi-Fi.

Connect sites

Join distant offices into one logical network.

Privacy

Hide traffic contents from the network in between.

Access control

Only authenticated devices get inside the tunnel.

05

Purpose: SAN

A network built for one job only — moving storage, at speed

Storage Area Network

SAN vs LAN

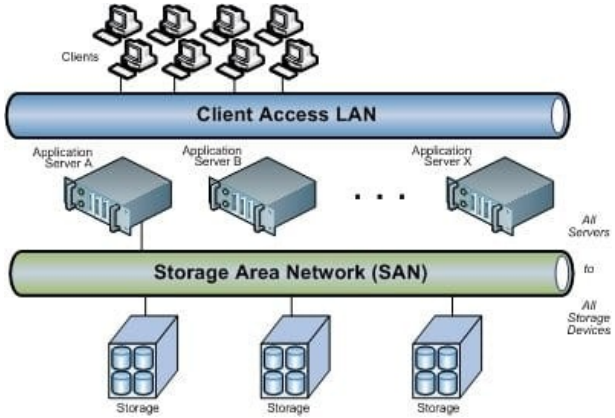
SAN vs NAS

Storage Area Network (SAN)

A separate, high-speed network dedicated entirely to storage

SAN

A SAN is a dedicated high-speed network that connects servers to shared pools of storage. Its whole reason to exist is to keep heavy storage traffic off the ordinary LAN, so that both the storage and the everyday network can run fast without interfering with each other.



Very high speed

Fibre Channel, typically 8 to 32 Gbps, tuned for storage.

Block-level access

Servers see the storage as if it were a local disk.

Separate from the LAN

Storage traffic never competes with user traffic.

SAN vs a Conventional LAN

They look similar, but they exist for opposite reasons

Aspect	Conventional LAN	SAN
Purpose	General traffic: users, apps, printing, Internet	One job only: connect servers to storage
Who uses it	People and their everyday devices	Servers and storage arrays — no users
What travels	Files, web pages, email, video	Raw storage blocks, at the disk level
Technology	Ethernet and Wi-Fi	Fibre Channel (or iSCSI over Ethernet)
Speed focus	Good enough for many users	Extreme throughput and low latency
Access type	File-level (you open a file)	Block-level (the server sees a raw disk)
If it is slow	A web page loads slowly	Every server and database stalls

The one-line distinction: a LAN connects people to services; a SAN connects servers to disks. Different users, different traffic, different purpose.

One Common Confusion: SAN vs NAS

Both share storage, but in very different ways

SAN

A whole network dedicated to storage

- Block-level: the server sees a raw disk
- Runs on its own Fibre Channel network
- Very high performance, very high cost
- For data centres and large databases
- Complex to design and manage

NAS

A single storage box on the ordinary LAN

- File-level: you open shared files and folders
- Plugs into the normal Ethernet LAN
- Modest performance, low cost
- For homes, small offices, backups
- Simple: plug in and share

Memory hook: NAS is a shared drive on your existing network; a SAN is a separate network that exists purely for storage.

Putting It Together: Classify These

For each scenario, name the type by scope and ownership



A smartwatch streaming your heart rate to your phone

PAN

One person, a few metres, Bluetooth.



All of Rajarata University's faculty buildings on one network

CAN

Several buildings, one owner, campus fiber.



A lecturer securely marking from home over the university's system

Remote-access VPN

One user, encrypted, over the Internet.



A bank linking head office to overseas branches on rented links

WAN (an EPN)

Global scope, private, leased lines.



A data centre connecting its servers to disk arrays over Fibre Channel

SAN

Storage-only, separate high-speed network.



Your phone, TV, and smart bulb on the home router

HAN / LAN

One home, one owner, shared Internet.

The method every time: ask how far it reaches (scope) and who controls it (ownership). Those two answers land you on the right name.

Activity

Sarasavi Institute of Technology is a small IT training institute. It has just moved into one building in Anuradhapura and you have been asked to plan the network.

The building has three areas:

- Administration office: 5 PCs, connected by cable
- Computer Lab: 10 PCs, all in one room, connected by cable
- Staff area: no fixed PCs. About 8 staff laptops and phones need wireless access

You also know that:

- One fibre broadband line from an ISP arrives at a wall socket in the building
- Instructors pair their phones with wireless earbuds while teaching
- The institute plans to open a second branch in Kurunegala next year

Activity

	Situation	Network type	Why
A1	The 10 lab PCs and the switch they plug into, inside one room		
A2	An instructor's phone paired with wireless earbuds while teaching		
A3	Next year's link joining the Anuradhapura building to the Kurunegala branch		
A4	A lecturer at home who needs a secure, encrypted path into the institute network over the Internet		

06

Practical Activity

Model multi-site networks and classify them in Packet Tracer

- Two-site WAN
- Site-to-site VPN concept
- Classify the build

Task 1

Model a Two-Site WAN

Join two LANs across a simulated wide-area link

1. Build LAN A: two PCs on a switch, on the 192.168.1.0/24 network.
2. Build LAN B: two PCs on a second switch, on the 192.168.2.0/24 network.
3. Add two routers, one for each site, and connect each router to its switch.
4. Connect the two routers together with a Serial or Gigabit link — this stands in for the WAN.
5. Address the router interfaces and configure a static route on each toward the other site.
6. Ping from a PC in LAN A to a PC in LAN B to prove the WAN link carries traffic.

Success looks like

A PC in site A can ping a PC in site B across the router-to-router link.

If it goes wrong

Cross-site ping fails but each LAN works alone? The static routes are missing or point the wrong way.

Task 2

Classify Your Own Build

Turn the theory back onto the network you just made

1. Look at each site on its own. What scope is one site — PAN, LAN, or CAN?
2. Now look at the whole build. What does the router-to-router link make it?
3. Write down the scope classification for the single site and for the whole thing.
4. Identify which parts you would own, and which you would rent from a carrier in reality.
5. Mark where a site-to-site VPN would sit if the link were the public Internet.
6. Note one change that would turn this WAN into an EPN.

Success looks like

You can label each part of your topology with the correct network type.

If it goes wrong

Remember: a single site is a LAN; two owned sites over a private link form a small private WAN, effectively an EPN.

Summary — Key Takeaways

Everything we covered in Week 3

Two axes	Every network is classified by geographic scope (how far it reaches) and by ownership (who controls it).
Scope tiers	PAN (personal) → HAN/LAN (a home or building) → CAN (a campus) → MAN (a city) → WAN (global). Speed falls as reach grows.
EPN	An enterprise private network: one company's own private WAN across all its sites. Secure, controlled, and expensive.
VPN	A secure encrypted tunnel across the public Internet. It gives EPN-like privacy at Internet-like cost, for remote access or site-to-site links.
SAN	A separate high-speed network dedicated to storage. It connects servers to disks at block level, keeping storage traffic off the LAN.
The method	To classify any scenario, ask two questions: how far does it reach, and who owns it?

Coming Up in Week 4

The OSI Reference Model

So far we have described networks by what they are and how big they are. Next week we look at how they are organised internally — the seven-layer model that explains how any two devices, anywhere, manage to understand each other.

The Seven Layers

From Physical up to Application — what each layer is responsible for.

Encapsulation

How data gains and sheds headers as it moves down and up the stack.

OSI vs TCP/IP

How the teaching model maps onto the model the Internet actually uses.

Layer-by-Layer Faults

Using the model as a checklist to locate exactly where a network breaks.

To prepare

Revise

Which device works at which layer, from Week 2

Classify

Spot three networks around you and name each type

Look ahead

The OSI model ties together everything so far